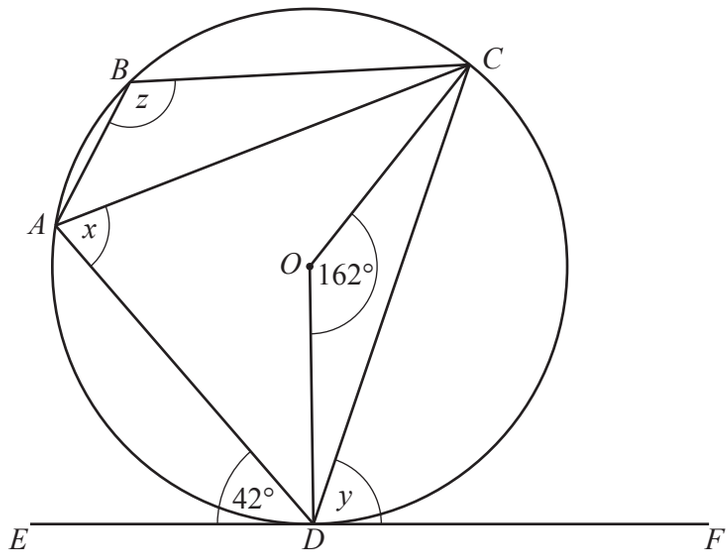
0580_w20_qp_43_Q5

5 (a)



NOT TO SCALE

A, *B*, *C* and *D* are points on the circle, centre *O*.
EF is a tangent to the circle at *D*.
Angle *ADE* = 42° and angle *COD* = 162°.

Find the following angles, giving reasons for each of your answers.

(i) Angle *x*

x = ... because .....

..... [2]

(ii) Angle *y*

y = ... because .....

..... [2]

(iii) Angle *z*

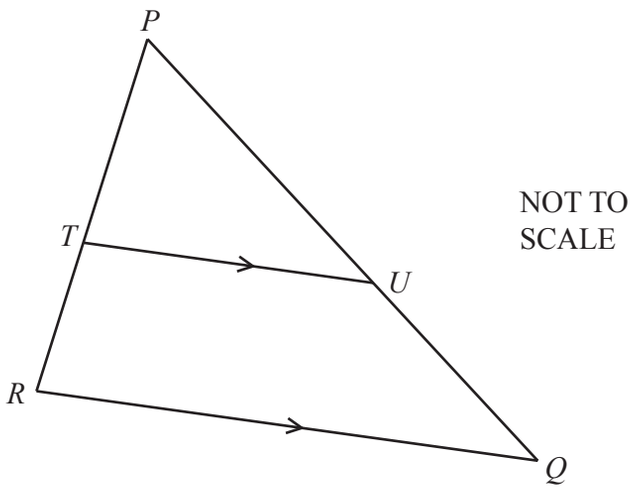
z = ... because .....

.....

..... [3]

0580_w20_qp_43_Q5

(b)



PQR is a triangle.
T is a point on *PR* and *U* is a point on *PQ*.
RQ is parallel to *TU*.

(i) Explain why triangle *PQR* is similar to triangle *PUT*.
Give a reason for each statement you make.

.....

.....

.....

..... [3]

(ii) $PT : TR = 4 : 3$

(a) Find the ratio $PU : PQ$.

..... : ..... [1]

(b) The area of triangle *PUT* is $20\,\text{cm}^2$.

Find the area of the quadrilateral *QRTU*.

..... cm^2 [3]